GStar Individual Project Guideline

The **GStar Program** emphasizes **practical innovations** and **technical impact**. Each student is expected to complete a **personal project** with two parts. **Part 1** serves as the **technical proposal** that lays the groundwork for **Part 2**. While you may **pivot or expand** your direction in Part 2, the proposal is your chance to **establish a strong foundation**. It will also be reviewed by TAs and lecturers, who will provide **feedback to guide your final project**.

Your immediate next assignment is the Project Proposal (Part 1)

1. Submission Details

1.1. [Current Focus] Part 1: Project Proposal

- Due Date: Sunday, October 4, 11:59pm PT
- **Technical Report:** Length: 1-2 pages
- Structure:
    1. Problem Description & Current Limitations
    2. Technical Proposal (central part): Technical approach, Evaluation methods, Compute requests
    3. Brief Literature Review on Related Problems
- *The theme for the problem can be found in section 2.*

1.2. [Future] Part 2: Final Submission

- Due Date: End of week 6
- Full written report
- GitHub repository (with code + documentation)
- Presentation slides + Demo

After the week 6 milestones, you may:

- Team up with others to expand your project, or
- Pivot into a new but related direction
- The goal is to build an advanced LLM or similar experience **in a team setting.**

2. Theme: *Advanced LLMs in a Specific Domain*

The theme is **intentionally open-ended** to maximize creativity and exploration.

Projects must be:

- **Demo-able within 6 weeks**
- **Practical** (clear use case and implementation)
- Technically challenging - The aim is for you to gain practical exposure to:
    - Post-training methods (SFT / RL)
    - OR agentic flows
    - OR data/compute efficiency wins
    - OR novel scalable techniques, algorithm changes
- While solving economically interesting tasks

You may advance LLMs in **any novel direction** or propose a **new benchmark.**

2.1. Suggested Project Directions

Here are **seed ideas** to inspire your proposal:

Tldr: We are seeking innovations in research or applications. The major theme includes long-horizon tasks, domain-specific tasks, or efficiency/novel research.

Suggestion: start by clearly identifying the problem you want to address: is it primarily an application-driven challenge or a fundamental research problem? Consider which aspect you want to focus on - whether it is exploring new architecture techniques, improving compute or data efficiency, or addressing limitations in current models. You should also think carefully about the specific

tasks or domains you want to target and whether they are well-established or require new formulations. Finally, decide how you will measure success: will you rely on existing benchmarks, design new benchmarks, or human evaluation?

The three seeds below are not mutually exclusive; they are some seed questions for you to think about different problems

[Seed 1] Domain specific tasks - Post-training (SFT/RL)

1. **Example 1: Deep Research**
    a. Context: Researching information is a highly economical task. Also, we should teach the model to provide citations where applicable, to support groundedness.
    b. Specialize models in **information retrieval and synthesis**
    c. **Reference works**: *OpenAI Deep Research*, *Gemini Deep Research*
2. **Example 2: API Environments.**
    a. Context: In the future, AI models are expected to use many tools. Hence, teaching the models to use tools effectively is a practical and critical problem.
    b. Train models to interact effectively with many APIs (e.g., reading files, integrating with YouTube, handling images).
    c. **Reference works**: *ToolFormer*, *ReAct*
3. **Domain-Specific Models**
    a. Context: General-purpose models are incredibly powerful, but there are occasions where you need powerful specialized models - like coding assistants, Math solvers, Spreadsheet/data analysis models, computer-use. Ideally a general-purpose model can do all of the above, but having specialized models are also a good differentiator
    b. **Reference works**: *OpenAI Codex*, *Qwen3-Coder*

There are a lot of other domain-specific tasks we can address.

[Seed 2] Test-time compute and Agentic workflows

We have observed that increased test-time computation is associated with improved performance. Understanding how to leverage test-time computation effectively is an open-ended research question. You should focus on specific problems to address, such as challenging/ long-horizon/ complex tasks. Test time computation can be achieved in many ways, via SFT, RL, or LMPs. Teaching the model to utilize more tokens is appreciated! This is intentionally open-ended for exploration, but ideally, your test-time-compute approach should be generalizable and enhance current LLM performance.

Reference works: *Gemini Deep Think*, *GPT-5 Pro*, *Grok 4 Heavy*, *AlphaEvolve*

[Seed 3] Architectural / Algorithm / Data changes

Even though LLMs are improving at a rapid rate, methods for improving **compute or data efficiencies** are still heavily appreciated. Optimizations across the stack allow us to train more efficient models or faster inference. Or even stability methods to help better model training.

Reference works: *FlashAttention*, *DeepSeek MLA*, *Muon* + *Kimi K2*.

2.2. Not Suggested: Avoid overly trivial projects, such as:

1. “Chat with PDF” (already zero-shot achievable)
2. A simple RAG Q&A bot without novel contributions
3. Non-LLM projects

3. Rewards - Recognition for Top Projects

1. Opportunities to **present at upcoming NTI conferences to industry leaders**
2. Additional **mentorship from the NTI leadership team** to expand your projects - with deep AI and product experience from frontier labs and leading startups

3. Potential **compute, research & industry support + publications opportunities** (depends on the scope of the work)

##

[Oct 3rd, 2026 Addition]

Appendix: Emerging Directions in AI

Given recent developments in artificial intelligence, we want to highlight a few ADDITIONAL directions that deserve more attention. Several already appear in the original guideline: data efficiency, post-training, agentic workflows, and domain-specific models. Recent work gives us more concrete ways to explore them.

We can think about these directions in terms of capabilities and alignment: making models more useful, and keeping their behavior reliable as they gain autonomy. The goal remains a practical, technically challenging project that is demo-able within six weeks.

1. Capabilities: How models learn

Data / Environment: Synthetic data is becoming a practical way to teach complex behaviors. Kimi K2 generates and filters tool-use interactions across many domains; EigenData pairs multi-turn dialogues with executable checkers. Work such as Kimi K2 and Zhipu AI's GLM-4.5 also highlights learning through interaction.

We think the interesting question is which data actually helps the model improve with clear ablations, and **we encourage you to think about scalable methods.**

Algorithms: Better capabilities should also become cheaper to train and use. DAPO studies sampling and optimization for reasoning RL, while slime supports asynchronous agent rollouts. These provide starting points for improving both algorithms and training infrastructure. Compare quality, time, and token usage at the same compute budget. **Architecture, algorithms, and optimizer improvements are always welcome!**

2. Capabilities: How models solve problems - multi-agent

Multi-agent systems extend the theme of test-time computation - inspired by promising multiagent systems results from OpenAI's Navier–Stokes work, which describes coordinating groups of agents.

We think how to make these systems work best remains an open question. What does it look like? When does collaboration outperform independent attempts at the same budget? How should agents work together? All remains open for exploration!

3. Capabilities: Going deeper into a vertical (Bio, Legal, etc.)

We still see room for domain-specific models and workflows. OpenAI's GPT-Rosalind is built for life sciences research. Anthropic's discovery of an enzyme system with CRISPR-like repeats offers another recent example of AI-assisted research followed by human laboratory work. In law, Google's Gemini 4 Argon achieved a 19.6% task pass rate on Harvey's Legal Agent Benchmark. **The benefits between generalized models vs industry-specific models remain open for discussion.**

For a six-week project, choose a narrow bottleneck: scientific evidence synthesis, a specialist model, or a reproducible analysis workflow. Use credible data and expert feedback where possible, and show what your approach improves over a strong general-purpose baseline.

A well-designed benchmark can itself be a meaningful contribution. As agents take on longer tasks, we need to evaluate whether they can use information, make sound decisions, and deliver useful work, beyond answering exam questions. Harvey's BigLaw Bench and Legal Agent Benchmark offer examples of domain-specific tasks and rubrics; the latter evaluates work products built from client materials. **You could develop a specialized benchmark focused on HIGH-VALUE knowledge work (finance, legal, office-work).**

4. Alignment: Teaching models why, not only what

Recent incidents make alignment science urgent! During tests with reduced safeguards, OpenAI models bypassed internet restrictions and compromised external systems. In misconfigured Anthropic tests without production cyber safeguards, Claude accessed third-party systems and uploaded a malicious PyPI package. Google confirmed that Gemini accessed three companies' systems during testing.

How do we teach models to recognize harm and stop, even when a task rewards continuing? Claude's Constitution and Teaching Claude Why offer starting points for teaching the principles behind good behavior. Test honesty,

respect for permissions, and acceptance of correction alongside task completion in isolated environments.

5. Alignment: Understanding human values

Does a model understand human values, and how could we tell? We think this goes beyond following instructions: values can conflict, and people may disagree about what a good outcome looks like. How should we train models to handle these differences and ask for guidance when needed? This is a very open-ended and nuanced question.

You can choose a setting and study how models respond to competing priorities. Experimenting with human feedback, explicit principles, or diverse perspectives, then evaluate behavior in unfamiliar situations.

6. Alignment: Human–AI interaction

If models become superhuman in some capabilities, what should their relationship with people look like? We think human–AI interaction remains an open question: when should a model lead, explain, challenge, or defer, and how do we preserve human judgment and agency? Can we still align models in that case - superalignment? Weak-to-strong generalization proposed the initial thought process, which slowly becomes more relevant as AI systems become more capable.

As Dr. Thang and Dr. Hung discussed - a model that solves a problem well may not help humans learn. Propose different approaches and discuss how to train and measure the AI impact!

To recap

These directions can be combined. Start with a problem you care about, identify what current approaches are missing, and make one contribution you can demonstrate convincingly.

Deadlines & Presentation Schedule

For Most Students

- **Submission deadline:** Tuesday, October 27, 2026, at 11:59 PM PT
- **Presentation sessions:**
    - Wednesday, October 28, 2026 | 7:00 - 9:00 AM PT
    - Friday, October 30, 2026 | 6:00 - 8:00 PM PT

For Late Joiners

- **Submission deadline:** Thursday, October 29, 2026, at 11:59 PM PT
- **Presentation session:** Friday, October 30, 2026 | 6:00 - 8:00 PM PT

Presentation Format & Logistics

Each student will have **4 minutes in total**:

- **2 minutes - Project Presentation:** Introduce your project, explain your technical approach, and highlight your implementation progress, results, and key findings.
- **2 minutes - Feedback & Q&A:** Faculty and TAs will ask questions and provide brief feedback.

To ensure that all students have sufficient time, please follow these guidelines:

- **Time limit:** Strictly adhere to the 2-minute presentation limit. We will keep time and may need to stop presentations that exceed the limit.
- **Slide limit:** Prepare a maximum of **2 slides**.
- **Shared slide deck:** All presentations will use one shared slide deck managed by the GStar team. You do not need to share your screen.
- **Be ready to present:** Please be ready when your name is called to ensure smooth transitions between presentations.
- **Feedback:** Due to time constraints, live feedback will be brief. Additional feedback may be shared afterward.

Guidelines for Revision & Submission

1. Revise and Finalize Your Project

Incorporate feedback from your proposal presentation and any changes or improvements you have made since then. It is completely fine if your project has evolved from your original proposal.

2. Prepare Your Presentation Slides

Prepare a maximum of **2 slides** and add them directly to the Master Presentation Slide Deck.

Please do not create a separate slide deck. We will use one shared deck throughout the presentations to ensure smooth transitions.

3. Focus on Your Approach and Results

Your slides should clearly communicate:

- Project idea and objectives
- Technical approach
- Implementation progress
- Results and key findings

Keep your slides concise, well-structured, and easy to follow.

4. Review Previous Examples

For inspiration, refer to the GStar 2025 Individual Project Presentation Slides to see how previous students presented their work.

5. Submit Your Final Write-up on E-learning

Please refer to the write-up template for guidance on the expected structure and content.

- **Submission platform:** E-learning
- **File naming format:** Your full name - Project Topic

Please review your file carefully before submitting, as **only one upload is allowed**.

6. Ask for Help if Needed

If you have any questions or need support while finalizing your project, please post in the **#individual-projects** Discord channel.

Good luck with your projects! We look forward to seeing what you have built and hearing about your work.